

Training-Free Visual Answer Confidence for Multimodal LLMs

Anonymous ACL submission

Abstract

Multimodal language models often answer visual questions with confidence even when the image evidence is partial, OCR-dependent, or ambiguous, making deployment hinge on whether to trust the model, call an external tool, or abstain. We study a training-free visual uncertainty estimator, VAC, that combines multi-prompt consistency, crop/zoom consistency, caption-VQA consistency, and OCR agreement into a calibrated confidence and action policy. All source conditions use the same hosted instruction-following vision service through a Responses-style API under a fixed 256-token answer budget with a 1024-token no-text retry; the SIEVES-style and VAC composites are offline policies over saved source rows and make no new model calls. On a completed 600-task matrix spanning OCRBench, VQAv2/AdVQA, and VizWiz VQA, VAC preserves the inherited direct-answer mean score of 0.682778 while reducing 10-bin ECE from 0.402222 to 0.130775. Multi-prompt consistency is the highest raw-score signal in this evaluation with mean score 0.702778, while source-specific signals expose different coverage, tool-use, and abstention tradeoffs. The results support a scoped claim: visual consistency signals are a useful training-free calibration layer for selective prediction, but this evaluation does not establish an official SIEVES reproduction, a raw accuracy gain for VAC, or multi-seed significance.

1 Introduction

Visual question answering systems are increasingly used as decision aids rather than only benchmark answer generators. In those settings, a model’s answer is not the whole interface: a system must decide whether to trust the answer, route the example to a specialized tool, or abstain. The difficulty is that multimodal LLMs can produce plausible answers when the visual evidence is weak, especially for OCR-heavy images, adversarial VQA variants, and accessibility images where ambiguity

is common. Prior work on multimodal calibration, selective VQA, and visual evidence scoring shows that raw confidence and self-reflection are brittle signals for these decisions (Chen et al., 2024; Padhi et al., 2025; Khan and Fu, 2024; Wiecek et al., 2025; Rodriguez and Rohrbach, 2026; Lan et al., 2024; Ye and Wen, 2025; Lan et al., 2025).

This paper asks whether a training-free confidence gate can be built from signals already available around a multimodal answer. The hypothesis is not that a new prompt wrapper should improve answer accuracy by itself. Instead, the paper tests a narrower mechanism: if several visual consistency probes agree with the direct answer, the system should be more willing to trust it; if the probes diverge, the system should lower confidence and route toward tool use or abstention. This framing follows the selective-prediction view of VQA confidence while avoiding task-specific fine-tuning or additional learned calibrators (Dancette et al., 2023; Sarkar et al., 2026; Shamsi, 2026).

We evaluate this idea with **Visual Answer Confidence** (VAC), a training-free composite over four source families: multi-prompt consistency, crop/zoom consistency, caption-VQA consistency, and OCR agreement. Figure 1 summarizes the policy. The method computes a confidence score and maps it to a trust/tool/abstain action; answer correctness is inherited from the underlying direct answer or source condition rather than rewritten after the fact.

Our evidence is a full 600-task benchmark-image matrix with 200 rows each from OCRBench, VQAv2/AdVQA, and VizWiz VQA. Across this matrix, VAC and the SIEVES-style proxy have the same mean score as direct answering, 0.682778, but VAC reduces 10-bin ECE from 0.402222 to 0.130775. Multi-prompt consistency has the highest observed mean score, 0.702778. These numbers motivate a calibrated, conservative story: VAC is useful for confidence and selective policy analy-

sis, while raw accuracy improvements and official SIEVES reproduction remain unsupported by the current evaluation.

The contributions are:

- A training-free visual confidence formulation that combines prompt, crop/zoom, caption, and OCR agreement signals for trust/tool/abstain policy.
- A 600-task evaluation over three real multimodal benchmark families, with direct, refusal-token, self-confidence, individual source-signal, SIEVES-style proxy, and VAC conditions.
- A calibration and selective-policy analysis that separates confidence quality, answer coverage, tool routing, and abstention behavior.

The contribution is a confidence-layer evaluation, not an answer-repair system. The same evidence that supports the calibration claim also shows why the method should be read this way: the full composite inherits the direct-answer score rather than rewriting wrong answers. This distinction is important for multimodal systems work because selective prediction can be useful even when answer generation is unchanged. A deployed assistant may still benefit from knowing which answers should be accepted immediately, which should be sent to an OCR or visual verification tool, and which should be withheld (Rodriguez and Rohrbach, 2026; Khanmohammadi et al., 2026; Cheng and Dong, 2026; Thanh, 2026). We therefore treat confidence, coverage, tool routing, and abstention as first-class outputs. The scope boundaries are also explicit: the SIEVES comparison is a SIEVES-style proxy over available source signals, and all intervals are task-resampling intervals over one completed run. These boundaries keep the central claim sharp: heterogeneous visual consistency checks can move a multimodal system from an overconfident answer-only interface toward an audited trust/tool/abstain policy.

2 Related Work

Multimodal calibration and uncertainty. Recent studies show that multimodal LLM confidence is often miscalibrated across VQA, OCR, hallucination, GUI, and medical settings (Chen et al., 2024; Padhi et al., 2025; Chen et al., 2026; Kumar et al., 2026; Khanmohammadi et al., 2026; Senoglu et al.,

2026; Pandey et al., 2025). These papers motivate calibration metrics such as ECE and selective risk, but many focus on a single uncertainty family, dataset slice, or learned/grounded calibration strategy. Our focus is narrower and operational: a training-free gate assembled from source consistency signals available during inference.

Consistency for selective VQA. Consistency-based methods test whether a model’s answer remains stable across prompts, views, or evidence transformations. Selective VQA work has shown that black-box response consistency can help identify unreliable answers (Khan and Fu, 2024; Wieczorek et al., 2025; Dancette et al., 2023; Sarkar et al., 2026; Shamsi, 2026). VAC extends that idea by combining multi-prompt, crop/zoom, caption-VQA, and OCR agreement signals in one calibrated action policy, then measuring both confidence calibration and coverage/risk behavior.

Abstention and evidence scoring. Abstention is useful when a model has insufficient knowledge or insufficient visual evidence. Recent visual evidence scoring work, including SIEVES, frames selective prediction as deciding which visual answers to accept (Rodriguez and Rohrbach, 2026). We compare against a SIEVES-style proxy because the official selector was not reproduced in this study. The paper therefore treats SIEVES as a proxy baseline and avoids claiming an official implementation.

VQA and OCR benchmark lineage. The evaluated tasks inherit from visual question answering and grounded visual reasoning benchmarks (Agrawal et al., 2015; Zhu et al., 2015; Andreas et al., 2015; Xu and Saenko, 2015; Wang et al., 2015; Zhang et al., 2015). OCRBench adds text-reading stressors for multimodal models (Liu et al., 2023; Shen et al., 2024; Nguyen et al., 2024; Han et al., 2020; Jain et al., 2021), while recent multimodal hallucination and OCR benchmarks emphasize that visual errors often arise from evidence extraction rather than language-only reasoning (He et al., 2025; Zhou et al., 2026; Jiang et al., 2026).

Positioning. The closest prior work motivates selective prediction, calibration, and visual evidence scoring, but the present experiment asks a different operational question. Instead of learning a new calibrator, sampling a large answer distribution, or requiring access to model internals, VAC asks whether a black-box inference harness can derive a useful confidence score from source agreement

checks that are natural for multimodal use: re-prompting, re-viewing, caption-mediated checking, and OCR agreement (Khan and Fu, 2024; Wic-zorek et al., 2025; Zou et al., 2026; Ge et al., 2024; Lin and Parikh, 2016). This makes the method cheaper and easier to bolt onto hosted systems, but it also constrains the expected gain. The right comparison is not whether VAC beats a trained verifier on every metric; it is whether the extra consistency probes expose actionable uncertainty without pre-tending to improve answer generation.

3 Method

VAC is a training-free confidence and action policy for a fixed multimodal answering backend. Given an image-question pair, the system first obtains a direct answer and then computes source consistency signals around that answer. The policy does not train a new model and does not rewrite the gold score. It changes the confidence and action attached to the answer.

Source signals. The multi-prompt signal measures whether prompt variants preserve the same answer. The crop/zoom signal checks whether local image views support the answer. The caption-VQA signal asks whether a caption-mediated question-answer view remains consistent with the direct response. The OCR agreement signal checks text evidence when the image contains readable text. These signals are chosen because they stress complementary failure modes: linguistic instability, visual locality, caption abstraction, and text extraction (Zou et al., 2026; Ge et al., 2024; Han et al., 2020; Shen et al., 2024).

Each signal is normalized as an agreement estimate rather than as an independent answer score. This keeps the confidence layer compatible with multiple VQA backends: a prompt-consistency signal can be computed from answer agreement, a crop/zoom signal from agreement under image views, a caption-VQA signal from agreement after caption mediation, and OCR agreement from text evidence when it is available. When a source is not informative for a row, the policy treats it as weaker evidence rather than as a new answer. This is why source signals can change confidence and action rates while leaving the inherited direct-answer correctness unchanged for the composite policies.

Composite score. The full composite uses a fixed weighted average over the direct-answer context

and the available consistency sources. The weights are 0.10 for direct-answer confidence, 0.15 for self-reported confidence, 0.15 for the refusal-token signal, 0.25 for multi-prompt consistency, 0.15 for crop/zoom consistency, 0.10 for caption-VQA consistency, and 0.10 for OCR agreement. The action rule abstains when the refusal source abstains or when confidence is at most 0.25, calls the tool for confidence below 0.67, and otherwise accepts the answer. The evaluation records the leave-one-signal sensitivity of this score: removing multi-prompt consistency changes mean confidence by 0.213333 and changes 349 actions, while removing crop/zoom changes mean confidence by 0.100417 and changes 83 actions. These weights should be read as a fixed analysis policy for this study, not as a learned optimum. The design goal is to test whether a simple heterogeneous combination is informative enough to warrant a larger learned or tuned selector in future work.

Calibration gate. The composite gate maps the source signals into a confidence score and an action. High-confidence examples are accepted; lower-confidence examples can be routed to a tool or abstained. In this evaluation, the tool action is evaluated as a policy decision rather than as an additional external correction run. This keeps the evidence honest: improvements in ECE and selective behavior are measured, but direct answer correctness remains the source of the raw score for the inherited composite conditions.

Why combine heterogeneous signals? The signals are intentionally heterogeneous because visual failures are not homogeneous. A prompt-consistency check can catch language instability but may miss that a small sign or object is outside the model’s attended region. Crop/zoom checks address that locality problem but can over-route to tools when the answer is globally obvious. Caption-VQA consistency compresses the image into a linguistic intermediate, which is useful for semantic agreement but can drop fine-grained text. OCR agreement targets exactly those text-heavy cases, but it should not dominate non-textual VQA. The composite gate is therefore designed as a routing layer over complementary evidence sources, not as a single universal confidence heuristic.

Action semantics. The trust action means the system accepts the model’s existing answer under the current confidence estimate. The tool action

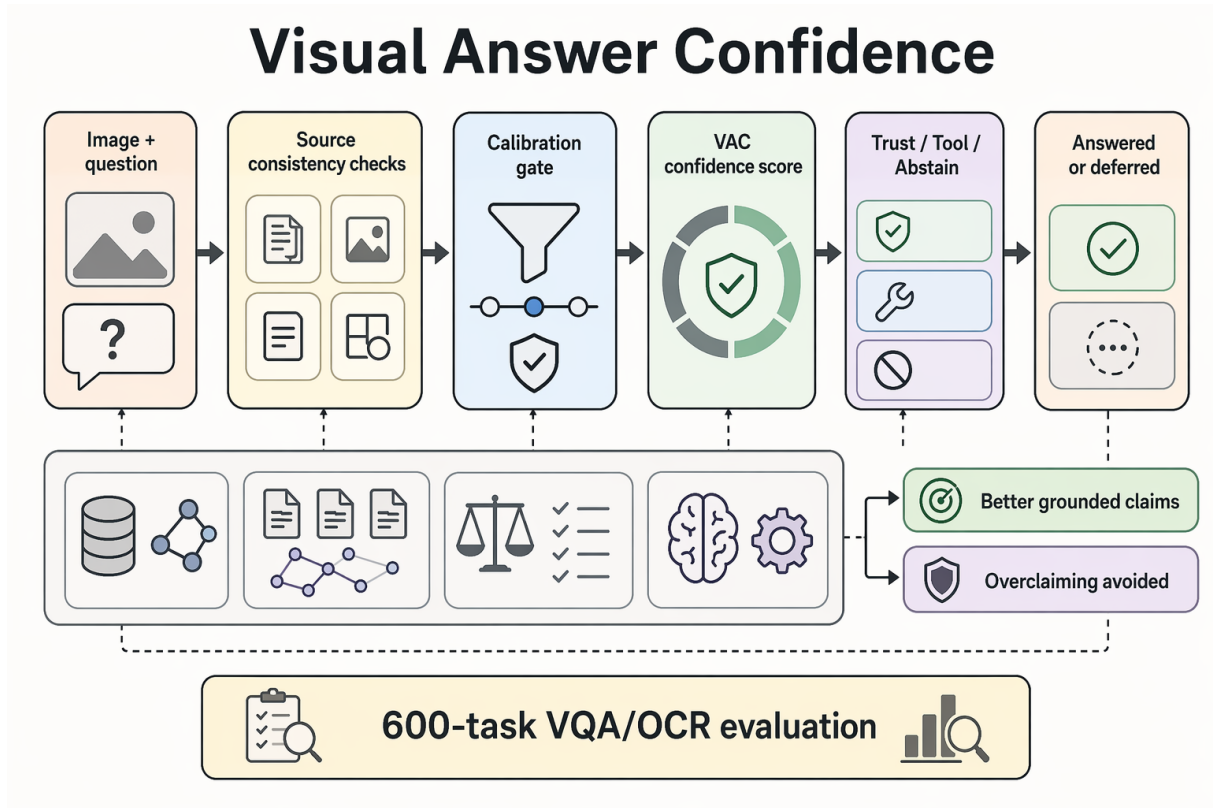


Figure 1: **VAC estimates visual answer confidence before action selection.** Visual source consistency checks feed a calibration gate that chooses whether to trust, call a tool, or abstain; the measured result is calibration/selective-policy evidence over 600 VQA/OCR tasks, not a raw accuracy gain.

means the answer should be checked by an external visual or OCR routine before being surfaced. The abstain action means the visual evidence is insufficient for a reliable answer. These actions are measured separately from answer correctness because a useful uncertainty policy can improve calibration and reduce overconfident answering without changing the underlying direct-answer score.

The policy is intentionally asymmetric. Accepting an answer exposes the user to wrong-answer risk, while tool routing spends extra computation or defers to a specialized routine, and abstention withholds an answer. The analysis therefore reports coverage, tool rate, abstain rate, accepted accuracy, and a utility proxy together. A confidence method that only lowers ECE by refusing most rows would not be operationally useful; conversely, a method that keeps high coverage but leaves calibration unchanged does not solve the trust decision. The main tables are organized to make these tradeoffs visible.

Baselines and proxy composites. We compare against direct answering, refusal-token policy, self-reported confidence, each individual consistency source, a SIEVES-style proxy, and the full VAC

composite. The SIEVES-style proxy uses available evidence axes in the benchmark evaluation and should not be read as the official SIEVES code path.

4 Experimental Setup

Benchmarks. The evaluation uses three real benchmark families with 200 full payload rows each: OCRBench, VQAv2/AdVQA, and VizWiz VQA. OCRBench stresses OCR and text recognition; VQAv2/AdVQA stresses general visual question answering under adversarial variants; VizWiz VQA stresses user-taken images and ambiguous visual evidence (Liu et al., 2023; Agrawal et al., 2015). Each condition is evaluated on the same 600-task intersection.

Backend and budget. All model-calling source rows use the same hosted instruction-following vision service through a Responses-style API. The evaluated service is treated as a black-box VLM backend because the provider interface available for this study does not expose a public model or backbone name. We therefore report the backend at the service level instead of substituting an unsup-

ported public model name. The source-condition budget is 200 rows per family; direct, refusal, self-confidence, multi-prompt, crop/zoom, caption-VQA, and OCR agreement rows use benchmark image inputs, a 256-token response attempt, a 1024-token no-text retry sequence, and a 180-second request timeout. Offline composites, including the SIEVES-style proxy and VAC, make no new model calls; they aggregate completed source rows. The evidence is a single completed evaluation, so bootstrap intervals are task-resampling intervals rather than seed dispersion.

Metrics. We report mean scored correctness, mean confidence, 10-bin ECE, accepted accuracy, answer coverage, call-tool rate, abstain rate, and a policy utility proxy. Mean score measures task correctness. ECE measures calibration of the confidence score. Coverage and risk measure whether the action policy accepts answers selectively rather than answering every row.

Evaluation scope. All headline claims use the 600-task benchmark evaluation described above. Pilot and development evaluations are excluded from the main claims. Each reported condition is scored on the same image-question identifiers across the three benchmark families, which prevents confidence comparisons from changing when difficult rows are silently removed. The numerical claims in the body are computed from the shared evaluation tables used to produce the reported figures and tables.

Row pairing. All main comparisons are made on the same 600 task identifiers. This matters because selective policies can appear stronger if difficult rows are silently removed. The evaluation keeps OCRBench, VQAv2/AdVQA, and VizWiz VQA balanced at 200 rows each and evaluates all reported conditions over that shared intersection. The paired structure is also why the paper reports task-resampling intervals for per-family score deltas. These intervals quantify row-level variation in one completed run; they are not independent random seeds.

Scoring and confidence bins. Correctness is the benchmark answer score recorded for each row. Confidence is the policy score attached to that row after a condition-specific source signal or composite is computed. ECE is computed with 10 bins from confidence and correctness. Accepted accuracy is the score among rows the policy accepts,

while answer coverage is the fraction of rows accepted. Tool and abstain rates account for the remaining policy mass. Reporting these quantities together prevents the paper from treating low ECE alone as sufficient: a confidence source can be well calibrated because it sends many cases to the tool action, which may or may not match a deployment budget (Chen et al., 2024; Padhi et al., 2025).

5 Results

Table 1 shows the main result. The raw-score best signal is multi-prompt consistency, which reaches 0.702778 mean score. VAC does not improve raw score over direct answering in this evaluation: both have mean score 0.682778. Its measured advantage is calibration and action policy. ECE drops from 0.402222 for direct confidence to 0.130775 for VAC.

Table 2 explains why the aggregate result should be read as a calibration result rather than as a universal accuracy improvement. On OCRBench, direct answering already scores 0.895000, leaving little room for an answer-only gain; VAC mainly changes confidence and routes 3.5% of rows to the tool action. On VQAv2/AdVQA, VAC keeps the direct mean score of 0.768333 while lowering ECE from 0.231667 to 0.093062. On VizWiz VQA, where direct answering scores only 0.385000 and abstains on 27.5% of rows, VAC does not solve the underlying visual ambiguity: it accepts fewer answers, routes 30.0% of rows to the tool action, and still has ECE 0.309493.

This family pattern is useful for method interpretation. The OCR-heavy family benefits from OCR agreement and caption-mediated checks because text evidence is often the decisive source. VQAv2/AdVQA benefits from multi-prompt stability and the composite’s ability to avoid treating every high-confidence direct answer as equally reliable. VizWiz is different: low-quality user-taken images and ambiguous questions produce uncertainty that source consistency can reveal but not necessarily resolve. In deployment terms, that is exactly the regime where a confidence gate should send more cases to verification or abstention.

The source signals behave differently across action dimensions. Crop/zoom, caption-VQA, and OCR agreement all call the tool frequently, while direct, refusal, self-confidence, and multi-prompt policies mostly answer or abstain. This supports the method design: VAC should be interpreted as

Benchmark/source	Method	Role	Score	ECE	Coverage	Tool/Abstain
All	Direct answer	Baseline	0.682778	0.402222	0.908333	0.000/0.092
All	SIEVES-style proxy	Proxy	0.682778	0.078972	0.748333	0.155/0.097
All	VAC full composite	VAC	0.682778	0.130775	0.785000	0.118/0.097
All	Multi-prompt consistency	Signal	0.702778	0.157222	0.898333	0.000/0.102
OCRBench	Direct answer	Baseline	0.895000	0.105000	1.000000	0.000/0.000
OCRBench	VAC full composite	VAC	0.895000	0.058500	0.965000	0.035/0.000
VQAv2/AdVQA	Direct answer	Baseline	0.768333	0.231667	1.000000	0.000/0.000
VQAv2/AdVQA	VAC full composite	VAC	0.768333	0.093062	0.970000	0.020/0.010
VizWiz VQA	Direct answer	Baseline	0.385000	0.870000	0.725000	0.000/0.275
VizWiz VQA	Multi-prompt consistency	Signal	0.436667	0.268333	0.700000	0.000/0.300
VizWiz VQA	VAC full composite	VAC	0.385000	0.309493	0.420000	0.300/0.280

Table 1: **The self-contained cross-benchmark matrix shows that multi-prompt consistency has the highest all-source mean score (0.702778), while VAC preserves direct-answer mean score (0.682778) and reduces ECE from 0.402222 to 0.130775.** Tool/Abstain reports call-tool and abstain rates. Hosted-source rows use the shared 600-task image benchmark with a 256-token answer attempt, a 1024-token no-text retry, and a 180-second timeout; the SIEVES-style and VAC composites aggregate saved source rows and make no new hosted-model calls.

Condition	Family	Score	Conf.	ECE	Coverage	Tool	Abstain
Direct answer	OCRBench	0.895000	1.000000	0.105000	1.000000	0.000000	0.000000
Direct answer	VizWiz VQA	0.385000	0.725000	0.870000	0.725000	0.000000	0.275000
Direct answer	VQAv2/AdVQA	0.768333	1.000000	0.231667	1.000000	0.000000	0.000000
Multi-prompt	OCRBench	0.895000	0.950000	0.055000	1.000000	0.000000	0.000000
Multi-prompt	VizWiz VQA	0.436667	0.685000	0.268333	0.700000	0.000000	0.300000
Multi-prompt	VQAv2/AdVQA	0.776667	0.925000	0.151667	0.995000	0.000000	0.005000
VAC full composite	OCRBench	0.895000	0.884770	0.058500	0.965000	0.035000	0.000000
VAC full composite	VizWiz VQA	0.385000	0.673085	0.309493	0.420000	0.300000	0.280000
VAC full composite	VQAv2/AdVQA	0.768333	0.861395	0.093062	0.970000	0.020000	0.010000

Table 2: **Per-family results show that the aggregate calibration gain is not a single-dataset artifact: VAC has lower ECE than direct answering on OCRBench (0.058500 vs. 0.105000) and VQAv2/AdVQA (0.093062 vs. 0.231667), while VizWiz remains the hardest family.**

an action policy over evidence signals rather than a universal answer selector (Chen et al., 2026; Kumar et al., 2026).

Calibration versus score. The headline numbers separate two axes that are often conflated. Multi-prompt consistency has the best observed mean score, 0.702778, but VAC has the paper’s central calibration behavior: it reduces direct-answer ECE by 0.271447 absolute while preserving the inherited mean score. Caption-VQA and OCR agreement have even lower aggregate ECE values, 0.074167 and 0.048056, but they also route more than half of all rows to the tool action. Those source-specific operating points are informative, but they are too conservative for a general answering policy if tool calls are expensive. The composite sits between answer-heavy and tool-heavy policies.

Per-family reading. The aggregate result hides a useful split. OCRBench is comparatively high scoring for direct answering and for the composite policies, while VizWiz VQA is the hardest family and produces the largest abstention and accepted-wrong mass. VQAv2/AdVQA sits between those regimes. This split is consistent with the paper’s design premise: the confidence gate is most valuable when the visual evidence can be missing, low quality, or dependent on text extraction. A single global answer confidence is therefore less informative than a confidence/action policy that can expose where answer coverage is being traded for verification (He et al., 2025; Jiang et al., 2026).

Task-resampling deltas. Table 3 reports representative per-family task-resampling intervals from the canonical results table. The intervals are paired by task within a family and compare each condi-

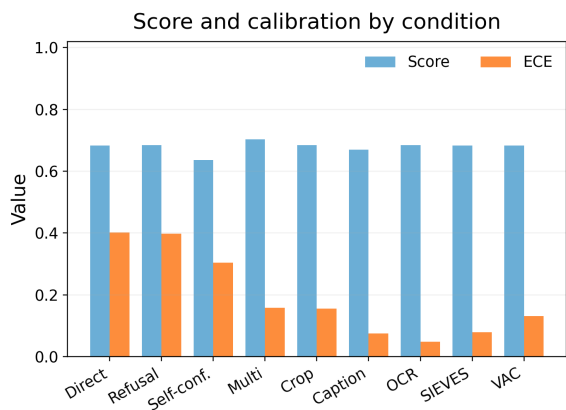


Figure 2: **Score and ECE separate raw correctness from calibration.** The plotted metrics show why the paper frames VAC as a calibration/selective-policy result rather than a raw accuracy gain.

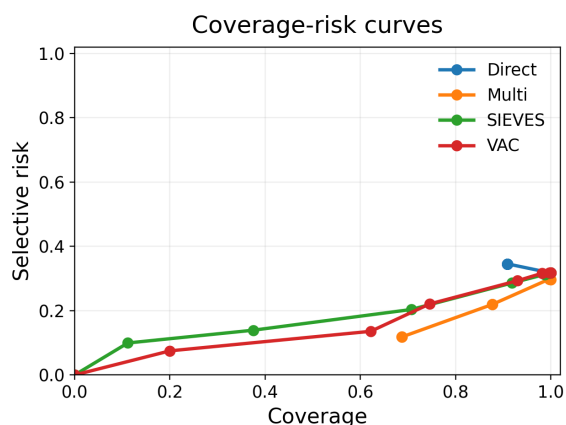


Figure 3: **Coverage and selective risk reveal the policy tradeoff.** The same 600-task benchmark evaluation is used to compare answering, tool routing, and abstention behavior under each confidence condition.

tion to direct answering. The VizWiz gains for multi-prompt and crop/zoom are positive under this resampling procedure, while crop/zoom is negative on OCRBench. This reinforces the source-heterogeneity argument: a signal that helps with one family can hurt or over-route another.

6 Analysis and Failure Cases

Selective behavior. Figure 3 reports the coverage/risk profile. The important pattern is not that one policy dominates all metrics, but that the confidence sources expose different operating points. OCR agreement has very low aggregate ECE but low answer coverage, while multi-prompt consistency retains high answer coverage and the best raw mean score. VAC trades between these regimes.

Signal sensitivity. The leave-one-signal diagnostic measures how much each source contributes to the VAC confidence/action policy. Zeroing the multi-prompt source changes mean confidence by 0.213333 and changes 349 of 600 actions, while zeroing the crop/zoom source changes mean confidence by 0.100417 and changes 83 actions. These are offline diagnostics over the evaluated VAC rows, not new model runs.

The action-change counts are more interpretable than the confidence deltas alone. Removing self confidence, refusal, or direct context changes more than 80 actions each, but the largest discrete shift comes from multi-prompt consistency. That means the prompt-stability signal is not merely a small calibration offset; it moves many examples across the trust/tool/abstain boundary. In contrast, caption-VQA and OCR agreement have smaller global action-change counts in the full composite, even though they can be strong on families where captions or text are central. This is another reason the paper does not reduce the method to a single source winner.

Failures. The failure taxonomy separates abstentions, accepted correct answers, accepted wrong answers, and tool deferrals. Direct answering accepts 204 wrong answers and abstains on 55 rows in the aggregate. This is the core deployment problem: high answer coverage is useful only if confidence is calibrated enough to flag when the visual evidence is weak.

The accepted-wrong mass is concentrated in the setting that motivates selective prediction. Direct answering has high confidence and high coverage, but its confidence score does not distinguish enough between easy OCRBench rows and ambiguous VizWiz rows. VAC reduces answer coverage from 0.908333 to 0.785000 and introduces a 0.118333 tool-routing rate. That tradeoff is not free: accepting fewer answers can frustrate users or increase latency. The value of the policy is that it makes the cost explicit. A system designer can decide whether the reduced overconfidence is worth the extra verification burden.

Qualitatively, the source families also fail for different reasons. Prompt consistency can agree on a wrong answer when every prompt follows the same spurious visual cue. Crop/zoom can lose context by isolating the wrong region. Caption-VQA can erase numerals, small objects, or spatial relations during the caption step. OCR agreement

Panel A: task-resampling score deltas vs. direct					Panel B: leave-one-source VAC sensitivity		
Cond.	Family	Δ	Low	High	Removed	Conf. Δ	Actions
Self-conf.	VizWiz	-0.083333	-0.128333	-0.038333	Direct	0.090833	80
Multi	VizWiz	0.051667	0.010000	0.098333	Self-conf.	0.121000	104
Crop	OCRBench	-0.055000	-0.090000	-0.020000	Refusal token	0.135500	114
Crop	VizWiz	0.071667	0.023333	0.125000	Multi-prompt	0.213333	349
Caption	OCRBench	-0.025000	-0.050000	0.000000	Crop/zoom	0.100417	83
OCR	VizWiz	0.013333	-0.025000	0.051667	OCR agree.	0.070917	47

Table 3: **Diagnostics show source-specific score movement and the dominant contribution of multi-prompt consistency: VizWiz deltas are positive for multi-prompt (0.051667) and crop/zoom (0.071667), while zeroing multi-prompt changes 349 of 600 VAC actions.** Panel A reports task-resampling intervals from one completed run rather than multi-seed significance; Panel B reports offline leave-one-source sensitivity over the VAC rows.

is useful when text is salient but can be irrelevant or misleading for non-textual scenes (Zhou et al., 2026; He et al., 2025). The composite is meant to average these failure modes rather than to declare any one signal universally reliable.

What the negative result means. The most important negative result is that VAC does not improve raw mean score in this evaluation. That outcome narrows, rather than invalidates, the claim: the method is a confidence and routing layer over a fixed answer source. A future answer-routing variant could use the source signals to change the answer itself, but the current evidence does not do so. The paper therefore treats the unchanged mean score as a constraint on interpretation and puts calibration, coverage, tool rate, and abstention behavior at the center of the contribution.

The negative result also clarifies the next research step. If the goal is raw accuracy, a future system must let the routing decision change the final answer, for example by invoking a real OCR tool, a higher-resolution crop verifier, or a second model on routed examples. That experiment would be a different method and would need to price tool cost and latency. The present study stops earlier: it shows that the confidence layer can identify and route uncertainty, but it does not claim that the routed branch has already been executed to repair answers.

Evidence sufficiency. The benchmark evaluation supports the paper’s claims about calibration, coverage, and signal diagnostics, but not broad robustness. The matrix is balanced across three benchmark families and nine reported conditions. However, it is still a single evaluation of one hosted vision service, with no official SIEVES selector and no repeated seeds. The paper therefore keeps

the main conclusion scoped to those measured conditions and uses the limitations section to state the missing evidence directly.

7 Conclusion

This study supports a scoped conclusion: training-free visual consistency signals can improve the calibration and action policy around multimodal LLM answers. In the 600-task benchmark evaluation, VAC preserves the inherited direct-answer mean score while substantially reducing ECE. The result is not a claim of raw accuracy improvement, official SIEVES reproduction, or multi-seed significance.

Limitations

The SIEVES comparison is a SIEVES-style proxy over available source signals, not the official SIEVES implementation. VAC inherits direct-answer correctness in this evaluation and therefore should not be described as improving raw accuracy. The reported bootstrap intervals are task-resampling intervals from one completed evaluation, not independent seed variation. Finally, the benchmarks cover three source families, but real deployments may require additional domains, tool costs, and human review policies.

Ethical Considerations

Confidence gates can reduce overconfident answering, but they can also create a false sense of safety if users treat abstention or tool-routing as a guarantee of correctness. The appropriate use is decision support: low confidence should trigger verification or human review, especially for accessibility, medical, or high-stakes images. This paper reports calibration and selective behavior from saved benchmark rows and does not claim autonomous reliability.

References

- Aishwarya Agrawal, Jiasen Lu, Stanislaw Antol, Margaret Mitchell, C. Lawrence Zitnick, Dhruv Batra, and Devi Parikh. 2015. [Vqa: Visual question answering](#). *Preprint*, arXiv:1505.00468.
- Jacob Andreas, Marcus Rohrbach, Trevor Darrell, and Dan Klein. 2015. [Neural module networks](#). *Preprint*, arXiv:1511.02799.
- Boqi Chen, Xudong Liu, Yunke Ao, and Jianing Qiu. 2026. [Revisiting greedy decoding for visual question answering: A calibration perspective](#). *Preprint*, arXiv:2604.23443.
- Zijun Chen, Wenbo Hu, Guande He, Zhijie Deng, Zheng Zhang, and Richang Hong. 2024. [Unveiling uncertainty: A deep dive into calibration and performance of multimodal large language models](#). *Preprint*, arXiv:2412.14660.
- Xueqi Cheng and Yushun Dong. 2026. [Latentrouter: Can we choose the right multimodal model before seeing its answer?](#) *Preprint*, arXiv:2605.11301.
- Corentin Dancette, Spencer Whitehead, Rishabh Maheshwary, Ramakrishna Vedantam, Stefan Scherer, Xinlei Chen, Matthieu Cord, and Marcus Rohrbach. 2023. [Improving selective visual question answering by learning from your peers](#). *Preprint*, arXiv:2306.08751.
- Yunhao Ge, Xiaohui Zeng, Jacob Samuel Huffman, Tsung-Yi Lin, Ming-Yu Liu, and Yin Cui. 2024. [Visual fact checker: Enabling high-fidelity detailed caption generation](#). *Preprint*, arXiv:2404.19752.
- Wei Han, Hantao Huang, and Tao Han. 2020. [Finding the evidence: Localization-aware answer prediction for text visual question answering](#). *Preprint*, arXiv:2010.02582.
- Haibin He, Maoyuan Ye, Jing Zhang, Xiantao Cai, Juhua Liu, Bo Du, and Dacheng Tao. 2025. [Reasoning-ocr: Can large multimodal models solve complex logical reasoning problems from ocr cues?](#) *Preprint*, arXiv:2505.12766.
- Aman Jain, Mayank Kothiyari, Vishwajeet Kumar, Preethi Jyothi, Ganesh Ramakrishnan, and Soumen Chakrabarti. 2021. [Select, substitute, search: A new benchmark for knowledge-augmented visual question answering](#). *Preprint*, arXiv:2103.05568.
- Yue Jiang, Xue Jiang, Lihua Zhang, Zhiqiang Wang, Yuhang Lu, Peng Wang, Bo Han, and Feng Zheng. 2026. [Mm-snowball: Evaluating and mitigating hallucination snowballing in multimodal multi-turn dialogue](#). *Preprint*, arXiv:2606.00622.
- Zaid Khan and Yun Fu. 2024. [Consistency and uncertainty: Identifying unreliable responses from black-box vision-language models for selective visual question answering](#). *Preprint*, arXiv:2404.10193.
- Reza Khanmohammadi, Kundan Thind, and Mohammad M. Ghassemi. 2026. [Calibrated triage, not autonomy: Confidence estimation for medical vision-language models](#). *Preprint*, arXiv:2606.15910.
- Divake Kumar, Sina Tayebati, Devashri Naik, Amanda Sofie Rios, Nilesh Ahuja, Omesh Tickoo, Ranganath Krishnan, and Amit Ranjan Trivedi. 2026. [Uncertainty quantification for computer-use agents: A benchmark across vision-language models and gui grounding datasets](#). *Preprint*, arXiv:2606.25760.
- Jian Lan, Diego Frassinelli, and Barbara Plank. 2024. [Mind the uncertainty in human disagreement: Evaluating discrepancies between model predictions and human responses in vqa](#). *Preprint*, arXiv:2410.02773.
- Jian Lan, Zhicheng Liu, Udo Schlegel, Raoyuan Zhao, Yihong Liu, Hinrich Schütze, Michael A. Hedderich, and Thomas Seidl. 2025. [Human uncertainty-aware data selection and automatic labeling in visual question answering](#). *Preprint*, arXiv:2510.11295.
- Xiao Lin and Devi Parikh. 2016. [Leveraging visual question answering for image-caption ranking](#). *Preprint*, arXiv:1605.01379.
- Yuliang Liu, Zhang Li, Mingxin Huang, Biao Yang, Wenwen Yu, Chunyuan Li, Xucheng Yin, and Chenglin Liu. 2023. [Ocrbench: On the hidden mystery of ocr in large multimodal models](#). *Preprint*, arXiv:2305.07895.
- Quan Van Nguyen, Dan Quang Tran, Huy Quang Pham, Thang Kien-Bao Nguyen, Nghia Hieu Nguyen, Kiet Van Nguyen, and Ngan Luu-Thuy Nguyen. 2024. [Vitetxvqa: A large-scale visual question answering dataset and a novel multimodal feature fusion method for vietnamese text comprehension in images](#). *Preprint*, arXiv:2404.10652.
- Trilok Padhi, Ramneet Kaur, Adam D. Cobb, Manoj Acharya, Anirban Roy, Colin Samplawski, Brian Matejek, and Alexander M. Berenbeim. 2025. [Calibrating uncertainty quantification of multi-modal llms using grounding](#). *Preprint*, arXiv:2505.03788.
- Ayush Pandey, Jai Bardhan, Ishita Jain, Ramya S Hebbalaguppe, Rohan Raju Dhanakshirur, and Lovekesh Vig. 2025. [Refine and align: Confidence calibration through multi-agent interaction in vqa](#). *Preprint*, arXiv:2511.11169.
- Hector G. Rodriguez and Marcus Rohrbach. 2026. [Sieves: Selective prediction generalizes through visual evidence scoring](#). *Preprint*, arXiv:2604.25855.
- Aditya Sarkar, Yi Li, Jiacheng Cheng, Shlok Mishra, and Nuno Vasconcelos. 2026. [Leveraging data to say no: Memory augmented plug-and-play selective prediction](#). *Preprint*, arXiv:2601.22570.
- Eren Senoglu, Federico Toschi, Nicolo Brunello, Andrea Sassella, and Mark James Carman. 2026.

Just how sure are you? improving verbalized uncertainty calibration in medical vqa. *Preprint*, arXiv:2606.27023.

Nina I. Shamsi. 2026. Density ridge selective prediction for llm and vlm hallucination detection under calibration label scarcity. *Preprint*, arXiv:2606.10198.

Zhixuan Shen, Haonan Luo, Sijia Li, and Tianrui Li. 2024. Adversarial training with ocr modality perturbation for scene-text visual question answering. *Preprint*, arXiv:2403.09288.

Duy Tran Thanh. 2026. Edges before embeddings: A confidence-aware blur gate for vision-language pipelines. *Preprint*, arXiv:2606.25838.

Peng Wang, Qi Wu, Chunhua Shen, Anton van den Hengel, and Anthony Dick. 2015. Explicit knowledge-based reasoning for visual question answering. *Preprint*, arXiv:1511.02570.

Tobias Jan Wiecek, Nathalie Daun, Mohammad Emtiyaz Khan, and Marcus Rohrbach. 2025. Variational visual question answering for uncertainty-aware selective prediction. *Preprint*, arXiv:2505.09591.

Huijuan Xu and Kate Saenko. 2015. Ask, attend and answer: Exploring question-guided spatial attention for visual question answering. *Preprint*, arXiv:1511.05234.

Yuanchang Ye and Weiyan Wen. 2025. Data-driven calibration of prediction sets in large vision-language models based on inductive conformal prediction. *Preprint*, arXiv:2504.17671.

Peng Zhang, Yash Goyal, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2015. Yin and yang: Balancing and answering binary visual questions. *Preprint*, arXiv:1511.05099.

Shizhe Zhou, Bohan Jia, Kai Wu, Yan Shen, Tongyun Li, Yuyang Wu, and Shaohui Lin. 2026. React-bench: A cause-driven benchmark for multimodal hallucination via systematic evaluation. *Preprint*, arXiv:2605.29579.

Yuke Zhu, Oliver Groth, Michael Bernstein, and Li Fei-Fei. 2015. Visual7w: Grounded question answering in images. *Preprint*, arXiv:1511.03416.

Hongjian Zou, Yue Ge, Qi Ding, Yixuan Liao, and Xiaoxin Chen. 2026. Caption first, vqa second: Knowledge density, not task format, drives multimodal scaling. *Preprint*, arXiv:2604.13054.

A Reproducibility Notes

The supplementary replay interface fixes the benchmark split, seed policy, evaluated hosted vision backend, and the 10-bin ECE/selective-prediction metrics used in the paper. It provides the evaluation tables needed to regenerate the reported score/ECE,

calibration/selective-risk, and signal-ablation summaries. Hosted source conditions use the shared 600-task image benchmark, a 256-token answer attempt, a 1024-token no-text retry, and a 180-second timeout; the multi-prompt signal uses three prompt calls under the same retry policy; and the SIEVES-style and VAC composites use saved source rows without new hosted-model calls. The supplementary material also includes the figure source used to render the method overview.

B Extended Evidence Map

The supplementary claim graph links six supported claims to the evaluation evidence. The evidence-gap notes record the three scoped limitations used in the main text: no official SIEVES selector, no VAC raw accuracy gain, and no multi-seed significance.